

VotePulse E-Voting Platform

Technical Specification: Module-by-Module Language, File Structure & Execution Workflow Report

Executive Summary: The VotePulse platform is an enterprise-grade, full-stack e-Voting and Election Management Progressive Web Application (PWA). It leverages modern asynchronous JavaScript (Node.js ES6+), React 19 component library, Express.js REST API layer, and Mongoose Object-Data Modeling over MongoDB. The application utilizes 256-bit SHA-256 cryptographic hashing alongside Caesar Cipher shift encryption to guarantee single-vote immutability, voter anonymity, and public auditability.

1. Comprehensive Module & Language Distribution Matrix

Module Name	Language / Tech	File Location Path	Key Responsibilities & Execution Logic
Voter Access & Voting Portal	JavaScript (JSX / React 19) HTML5 / CSS3	src/components/VoterPortal.jsx	Voter authentication, real-time Gmail OTP verification modal, live poll standings, single-vote ballot submission, Caesar Cipher receipt copy.
Candidate Command Center	JavaScript (JSX / React 19) Vanilla CSS	src/components/CandidatePortal.jsx	Candidate login & nomination registration, manifesto studio, live vote share analytics, campaign announcements, digital verified badge generation.
Administrator Console	JavaScript (JSX / React 19) Vanilla CSS	src/components/AdminConsole.jsx	Restricted admin auth, election lifecycle management (active/closed), candidate registration, real-time analytics tally, MongoDB visual inspector.
Ballot Audit & Verification	JavaScript (JSX / React 19)	src/components/BallotAuditTool.jsx	Public cryptographic verification portal. Accepts Caesar/SHA-256 hashes, decrypts shift cipher, verifies vote presence without breaking anonymity.
Application Shell & Router	JavaScript (JSX / React 19) HTML5	src/App.jsx index.html	Client-side SPA hash/pathname router (#voter, #candidate, #admin, #audit), theme manager, PWA install prompt banner, global user session state.
Navigation & Quick Route	JavaScript (JSX / React 19)	src/components/Navbar.jsx	Top navigation bar with active route pills, secret admin gate (triple-click logo / Ctrl+Shift+A), reactive Quick Admin/Voter switch button.
Backend REST API Server	Node.js (ES6+) Express.js	server.js	Express HTTP web server listening on port 3000. Handles authentication, vote casting, elections CRUD, admin metrics, and static asset serving.
Database Engine (NoSQL)	Node.js / Mongoose MongoDB BSON	database/mongo_db.js database/db.js	MongoDB ODM layer defining Mongoose schemas (User, Election, Candidate, Vote, GmailToken). Handles queries, document creation, and auto-seeding.
Email & OTP Dispatcher	Node.js / Nodemailer SMTP Protocol	utils/email.js	Gmail SMTP integration sending real 6-digit verification codes to voters' Gmail addresses using custom HTML email templates.
Cryptographic Security Engine	Node.js / Native Crypto	utils/cipher.js	Encapsulates 256-bit SHA-256 hashing and custom Caesar Cipher shift encryption/decryption functions for tamper-proof vote sealing.
Backend Request Validation	Node.js (CommonJS)	middleware/validation.js	Server-side middleware enforcing voter registration rules, 5-character password constraints, OTP formats, and vote payload validation.

Module Name	Language / Tech	File Location Path	Key Responsibilities & Execution Logic
Progressive Web App (PWA)	JavaScript / JSON Web App Manifest	sw.js manifest.json	Service worker script enabling offline caching (v106 cache buster), background fetch intercept, asset pre-caching, standalone app install.

2. Detailed Execution Workflows & Module Operations

A. Voter Authentication & Gmail OTP Verification Workflow

- Credentials Entry:** Voter inputs Voter ID / Gmail & Password (max 5 chars) in VoterPortal.jsx.
- Server Verification:** server.js invokes db.findUserByVoterId() and verifyUserPassword().
- OTP Generation & Email Dispatch:** createGmailToken() generates a 6-digit token in MongoDB. utils/email.js dispatches an HTML email via Gmail SMTP.
- Token Input:** Voter enters 6-digit code in the modal. verifyGmailToken() validates code and expiry, granting authenticated access.

B. Encrypted Vote Casting & Cryptographic Receipt Workflow

- Ballot Selection:** Authenticated voter selects candidate in VoterPortal.jsx.
- Single-Vote Verification:** Server checks db.hasVoted(election_id, voter_id) in MongoDB to prevent duplicate voting.
- Cryptographic Sealing:** Server encrypts payload with caesarCipherEncrypt(raw, 3) and generates sha256Hash(raw).
- Tally Update:** Candidate vote_count is atomically incremented by 1 in MongoDB.
- Receipt Issuance:** Voter receives a unique 256-bit cryptographic receipt hash for independent verification.

C. Candidate Command & Manifesto Studio Workflow

- Authentication & Registration:** Candidates log in via key or register nomination in CandidatePortal.jsx.
- Manifesto Management:** Candidates write/edit their campaign promises & slogans; saved to MongoDB and synced live to voter ballots.
- Real-Time Standings:** 3-second auto-refresh polling fetches live candidate tallies, rank (#1 Leader), and vote percentages.
- Verification Badge:** Generates digital candidate verification card with SHA-256 seal.

D. Administrator Master Control & Database Inspector

- Admin Gateway:** Authenticated via ADM-9999 credentials in AdminConsole.jsx.
- Election & Candidate Creation:** Admin creates new polls or registers candidates directly in MongoDB.
- Voter Management:** Displays registered voters list with deletion capabilities.
- Database Inspector UI:** Queries real-time collection counts (Users, Elections, Candidates, Votes, GmailTokens).

3. Technology Stack & Language Metrics

Layer	Languages / Tooling	Primary Dependency	Project Impact & Role
Frontend UI	JavaScript (ES6+), JSX, HTML5, CSS3	React 19, Lucide React	Interactive user interfaces, glassmorphism UI, real-time polling updates.
Backend API	JavaScript (Node.js CommonJS)	Express.js v4.22, Cors, Dotenv	RESTful endpoints, route security, static assets serving.
Database	NoSQL BSON Document Store	Mongoose v9.9, MongoDB	ACID compliance, document persistence, relational model mapping.
Security & Email	Node Native Crypto, SMTP	Nodemailer v9.0, SHA-256	Gmail OTP token delivery, Caesar Cipher shift ballot encryption.